

## **COMP634 - Assessment 3**

### **A Comparative Study of Neural Network Models for Part-of-Speech Tagging**

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## **1. Introduction**

Part-of-speech (POS) tagging is an NLP task. Each word in a sentence is assigned a label based on its grammatical function. This report compares two neural approaches for POS tagging: a Bidirectional Long Short-Term Memory (BiLSTM) model and a Transformer-based model. Both models are trained and evaluated under identical experimental conditions to assess their generalisation performance and to identify the most suitable architecture for this task.

## **2. Data Management and Experimental Protocol**

### **2.1 Dataset and Splitting Strategy**

The experiments used a POS tagging dataset in CoNLL-style format. In this format, tokens and their corresponding tags are listed line by line. Blank lines are left between sentences. There are a total of 3,914 sentences in the dataset. On an average, the sentences contain around 21 tokens. The pattern reflects a typical sequence labelling setting.

The dataset was split at the sentence level into training (80%), validation (10%), and test (10%) sets using a fixed random seed. This resulted in 3,131 training sentences, 391 validation sentences, and 392 test sentences. The validation set was used exclusively for model selection and early stopping. The test set was held out for final evaluation only. Sentence-level splitting preserves complete token-tag sequences. It also ensures a clean separation between training, tuning, and evaluation stages.

### **2.2 DataLoader and Pre-processing**

#### **2.2.1 Exploratory Analysis**

Before training, the dataset was checked to ensure data quality and to guide later design choices. All sentences were found to contain correctly aligned token-tag pairs. There were no mismatches present. Sentence lengths varied between 1 and 171 tokens. Most sentences were shorter than 40 tokens.

In total, the dataset contains 11,943 unique tokens. There are 33 part-of-speech tags. Some tags appear far more often than the others. High-frequency tags such as nouns and prepositions dominate the distribution, while several tags appear only rarely. This imbalance motivated the use of class-balanced evaluation metrics alongside accuracy. Numeric tokens and other irregular forms were retained to keep the data linguistically realistic, instead of applying aggressive normalisation.

#### **2.2.2 Vocabulary Construction and Encoding**

Vocabularies were constructed only on the training data. Two special tokens were used in encoding - <PAD> for handling different sentence lengths and <UNK> for words not seen during training. The final vocabulary had 10,577 tokens. For output labels, 34 tag indices were defined. This included a padding label, along with an inverse mapping for decoding.

Rates of previously unseen words were roughly 9.6% on the validation set and 8.4% on the test set. This means that most words were already seen during training. Only a small number of new words appeared.

### 2.2.3 Dataset and Batching Strategy

A custom PyTorch Dataset class was implemented to return encoded token sequences, encoded tag sequences, and original sequence lengths. A custom collation function pads sequences within each batch to the maximum length, generates boolean masks for valid tokens. It also retains sequence lengths for length-aware processing. Separate DataLoaders were used for training, validation, and testing. Shuffling was applied only to the training batches.

## 3. Neural Network Architectures and Training Components

### 3.1 Bidirectional LSTM Tagger

The BiLSTM model embeds tokens using a trainable embedding layer. It then processes sequences with a two-layer bidirectional LSTM. This design enables the model to capture contextual information from both preceding and following tokens. The resulting forward and backward hidden states are concatenated. They are then passed through a dropout layer. After this, a linear classifier produces token-level tag logits. Packed sequences are used to ensure that padded positions do not influence recurrent computations. This allows the model to work with sentences of different lengths.

The final BiLSTM uses the following - embedding dimension of 128, a hidden size of 128, and a dropout rate of 0.5. Earlier experiments with using a hidden size of 256 with a dropout rate of 0.3 caused the model to overfit. This prompted a reduction in model capacity to improve generalisation. This configuration was selected to balance expressive power with regularisation while retaining the sequential inductive bias required for effective POS tagging.

### 3.2 Transformer-Based Tagger

The Transformer model embeds tokens and adds learned positional embeddings to encode sequence order. Contextual representations are computed using three Transformer encoder layers with multi-head self-attention and feed-forward sublayers. Padding positions are masked during attention computation, and a linear classifier produces tag predictions. The model uses a 256-dimensional hidden representation, eight attention heads, and dropout 0.2.

### 3.3 Optimisation and Regularisation

Both models were trained using the AdamW optimiser with weight decay. Earlier experiments using higher learning rates resulted in unstable optimisation and overfitting. Final learning rates were set to  $1 \times 10^{-3}$  for the BiLSTM and  $1 \times 10^{-4}$  for the Transformer. Training employed cross-entropy loss with padding ignored and label smoothing (0.1). Gradients were clipped to a maximum norm of 1.0. Early stopping with a patience of three epochs was applied based on validation macro-averaged F1-score.

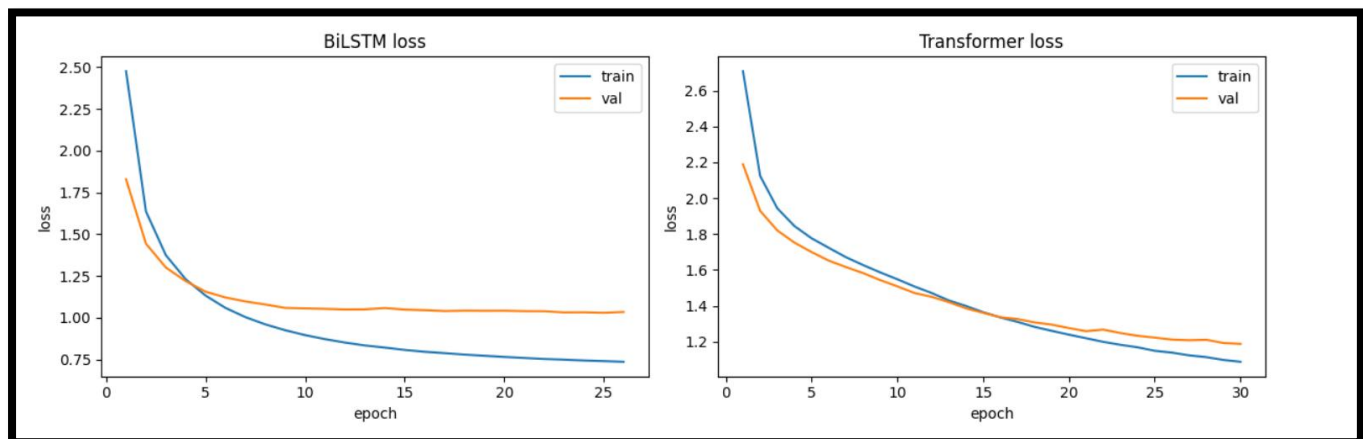
## 4. Results and Model Comparison

Model selection was based on validation macro-averaged F1-score. The score provides a balanced assessment across POS tags. The BiLSTM converged rapidly. It achieved a best validation macro-F1 of 0.826. The Transformer achieved a validation macro of 0.728. Validation results are shown in Table 1. Basis the results, the BiLSTM model was chosen as the best-performing model.

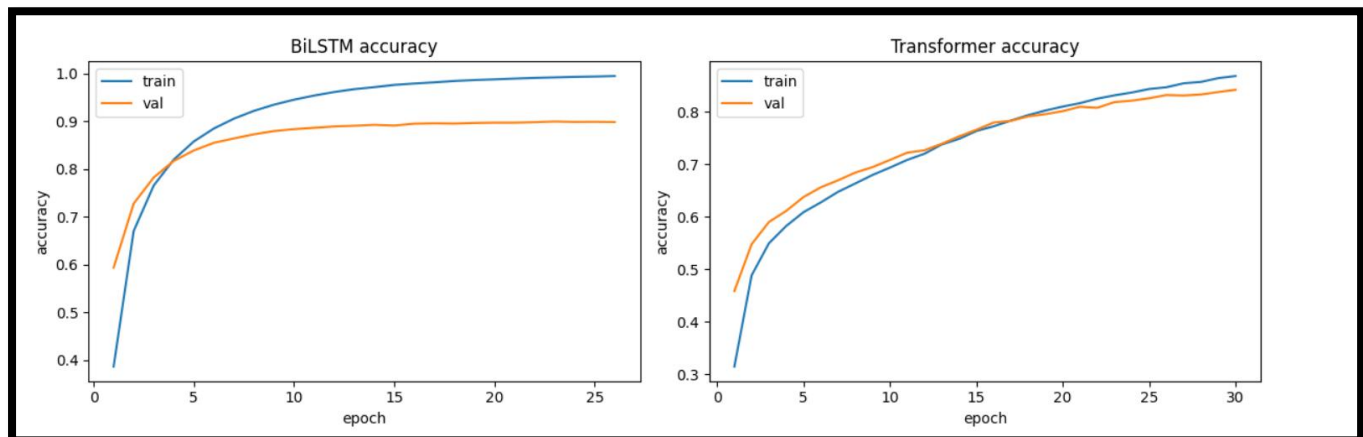
| Model       | Validation Macro-F1 |
|-------------|---------------------|
| BiLSTM      | <b>0.826</b>        |
| Transformer | 0.728               |

**Table 1: Comparison of validation performance**

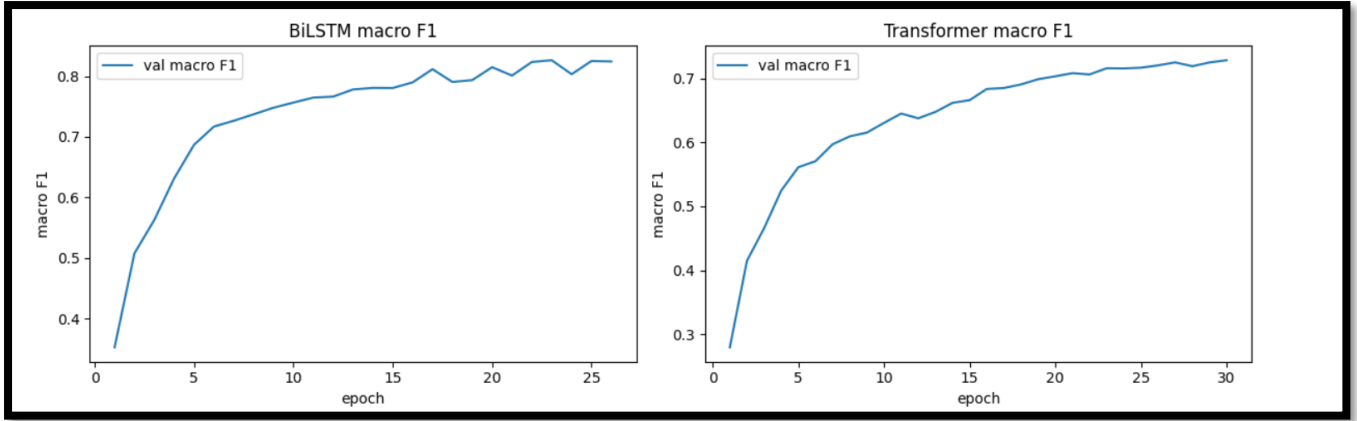
Training dynamics are shown in Figures 1-3. As shown in Figure 1, the BiLSTM converges more quickly than the Transformer. The gap between training and validation loss is also smaller. This suggests that regularisation is effective. Figure 2 shows that the BiLSTM reaches stable validation accuracy earlier during training. Figure 3 shows that the BiLSTM consistently achieves higher validation macro-F1 scores. This shows that the BiLSTM is the better model.



**Figure 1. Training and validation loss across epochs for the BiLSTM and Transformer models.**



**Figure 2. Training and validation accuracy across epochs for both models.**



**Figure 3. Validation macro-averaged F1-score across epochs for both models**

## 5. Evaluation on the Test Set

The models were evaluated using the test dataset. A range of complementary metrics were used. These include token-level accuracy, macro-averaged F1, weighted F1, and sentence-level exact match. The results are summarised in Table 2.

| Model         | Accuracy     | Macro F1     | Weighted F1  | Sentence EM  |
|---------------|--------------|--------------|--------------|--------------|
| <b>BiLSTM</b> | <b>0.910</b> | <b>0.831</b> | <b>0.911</b> | <b>0.225</b> |
| Transformer   | 0.851        | 0.769        | 0.848        | 0.110        |

**Table 2. Comparison of BiLSTM and Transformer performance on the held-out test set.**

The BiLSTM achieved an accuracy of 0.91. It also achieved a macro-F1 score of 0.83 and a weighted F1 score of 0.91. The sentence-level exact match score was 0.22. This indicated that roughly one in five test sentences were tagged perfectly. The metric is strict. It gives a clear indication of how well the model can produce fully correct tag sequences.

The Transformer on the other hand, achieved an accuracy of 0.85, macro-F1 of 0.77. The sentence-level exact match score was 0.11. The results indicate greater sensitivity to local tagging errors.

Figures 4 and Figure 5 display the confusion matrices of both models. As seen in Figure 4, the BiLSTM confusion matrix exhibits strong diagonal dominance. Most errors are occurring between syntactically similar tags. In contrast, Figure 5 shows that the Transformer produces more dispersed errors, particularly for lower-frequency tags. These observations corroborate the quantitative results and confirm the BiLSTM’s stronger generalisation.

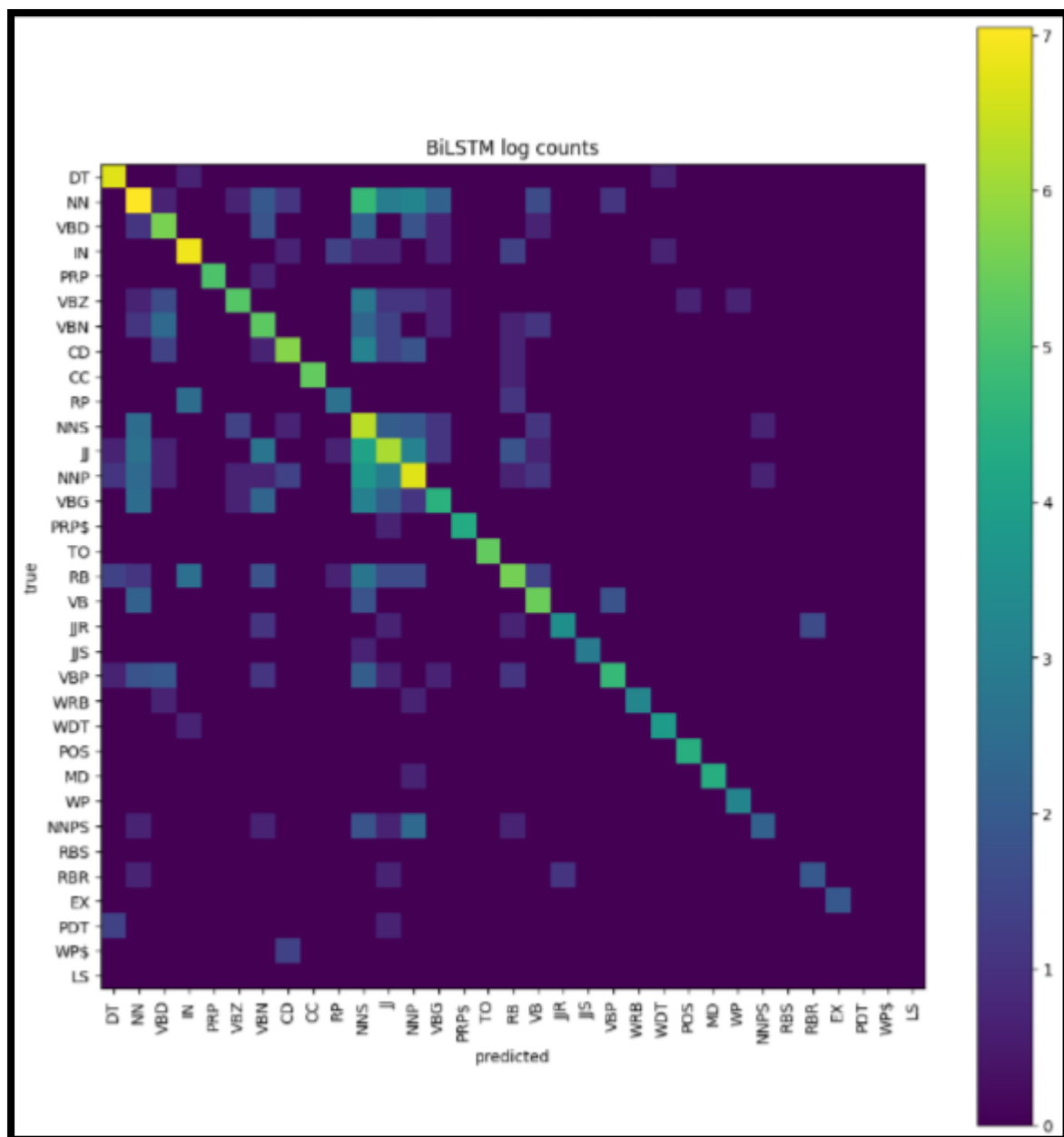


Figure 4. Log-scaled confusion matrix for the BiLSTM model on the test set.

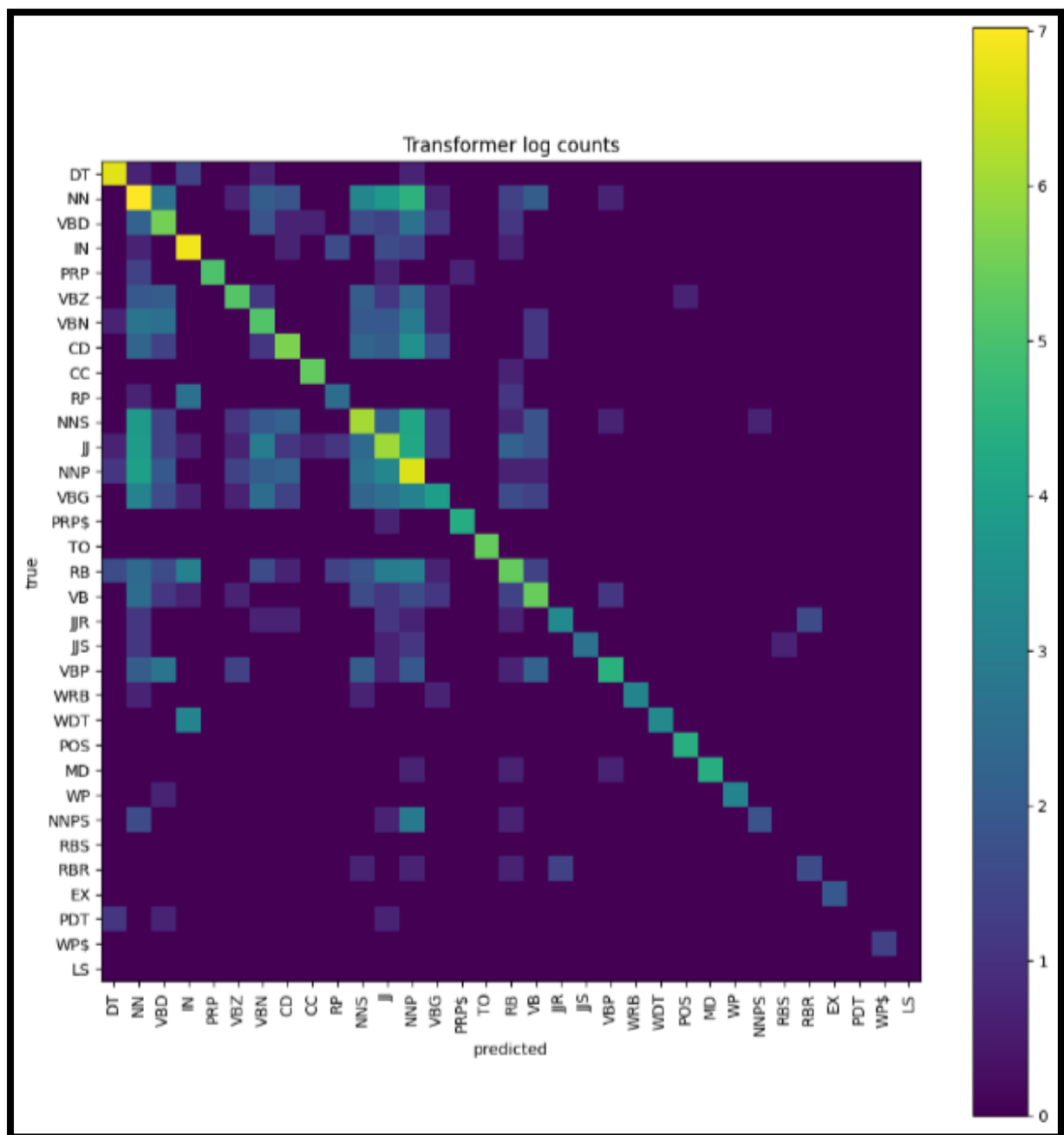


Figure 5. Log-scaled confusion matrix for the Transformer model on the test set.

## 6. Conclusion

This study compared a BiLSTM and a Transformer model for POS tagging. In both validation and testing, the BiLSTM showed more reliable results. This reflects the advantage of sequential inductive bias and targeted regularisation. While the Transformer demonstrated steady learning, it was less effective on this dataset. Overall, the results show that recurrent architectures remain well suited to sequence labelling tasks. This holds when the sentence lengths are moderate and the amount of training data is limited.